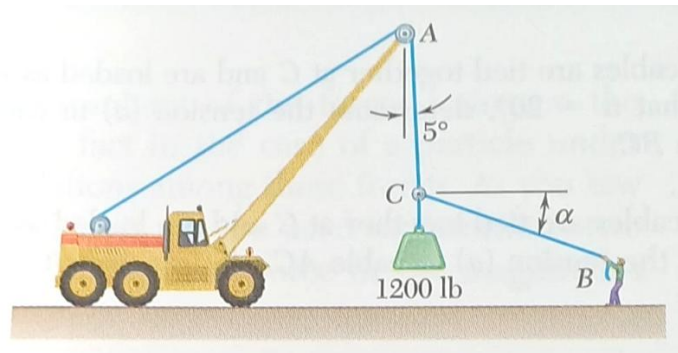


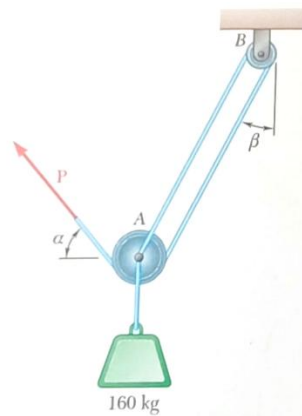
ME1103 Topical Practice Problems

Vector Forces and Moments

1) Given that $\alpha = 20^\circ$, find the tension in AC and BC. (1lb = 0.45kg)



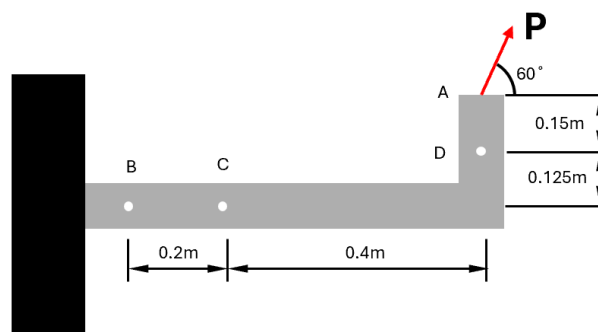
2) A 160kg load is supported by the rope and pulley arrangement shown.



a) If $\beta = 20^\circ$ is given, determine the magnitude and direction of force P.

b) If $\alpha = 40^\circ$ is given, determine β and the magnitude of P.

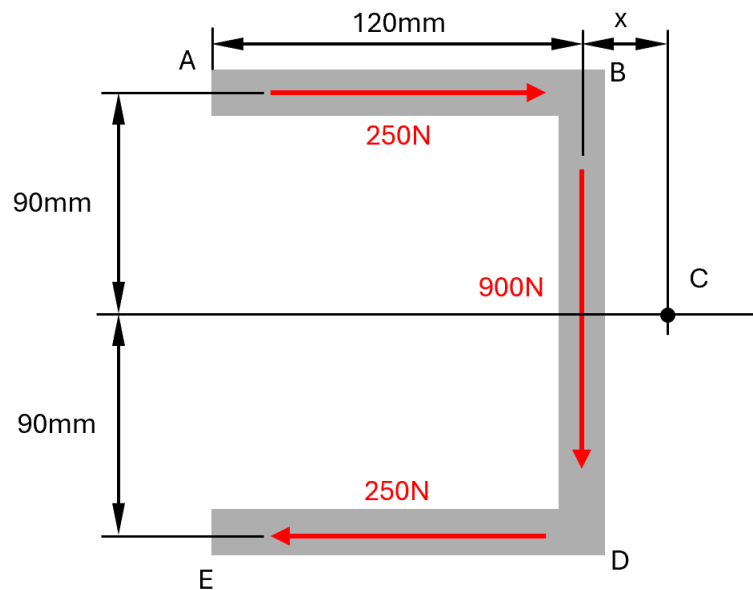
3) A 160N force P is applied at point A of a structural member. Replace P with



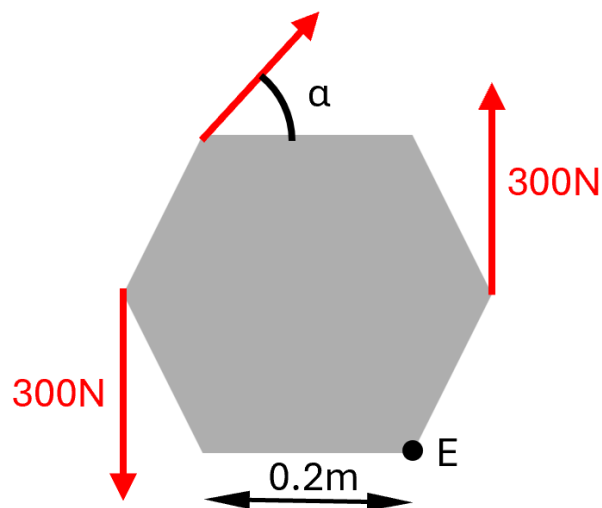
a) an equivalent force-couple system at C

b) an equivalent system of a vertical force at B and a second force at D

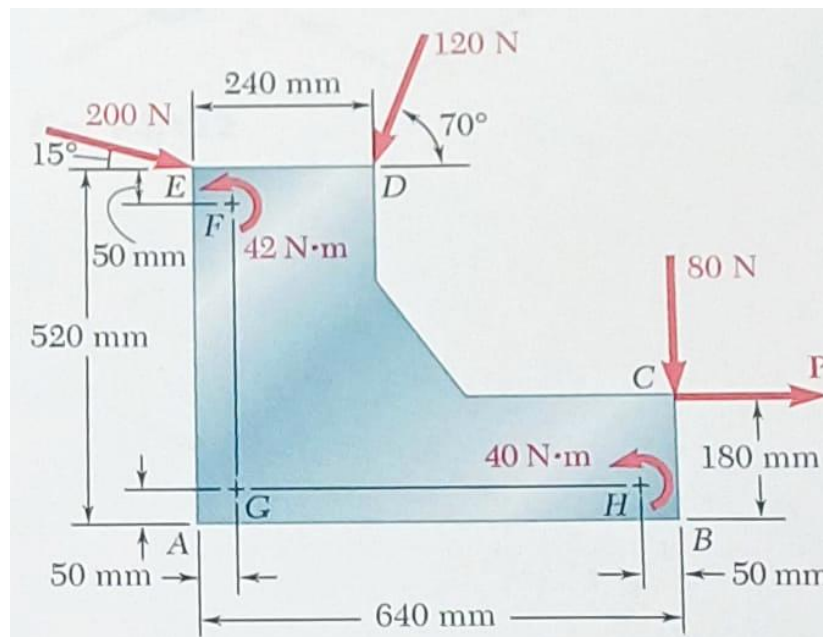
4) The shearing forces on the cross section of a steel channel can be represented by a 900N vertical force and two 250N horizontal forces as shown. Replace the forces with a single force at point C, and determine the distance x , of C to BD.



5) A regular hexagonal plate is acted upon the force P and the couple shown. Determine the magnitude and direction of the smallest force P for which the system can be replaced by a single force at E.



6) A machine component is subjected to the forces and couples shown. The component is to be held in place by a single rivet that can resist forces but not a couple.

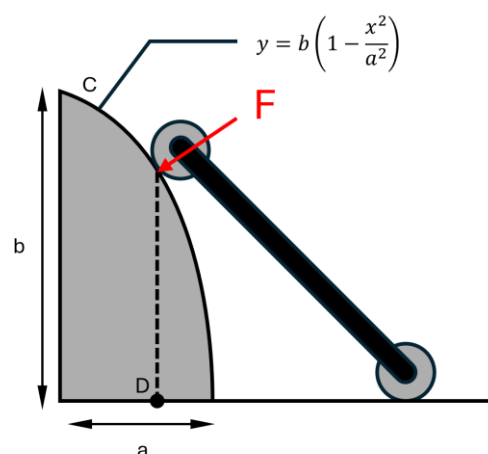


a) When $P = 0\text{ N}$, determine the location of the rivet hole if it is to be located on

(i) line FG

(ii) line GH

7) A cam follower AB rolls along the surface of cam surface C. It exerts a constant force F perpendicular to the surface.

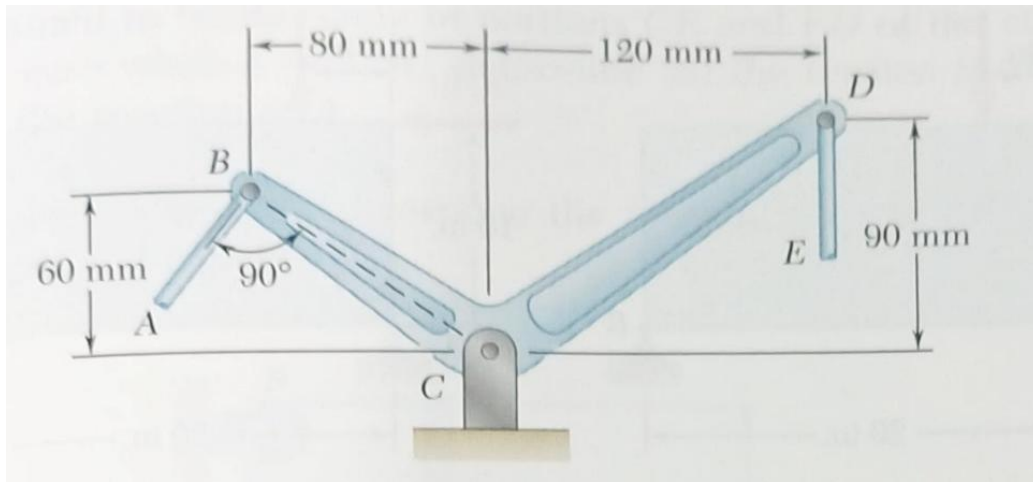


a) Replace F with an equivalent force-couple system at point D shown.

b) For $a = 1\text{ m}$ and $b = 2\text{ m}$, determine the value of x for which the moment of the equivalent force couple at D is maximum.

Static Rigid Bodies

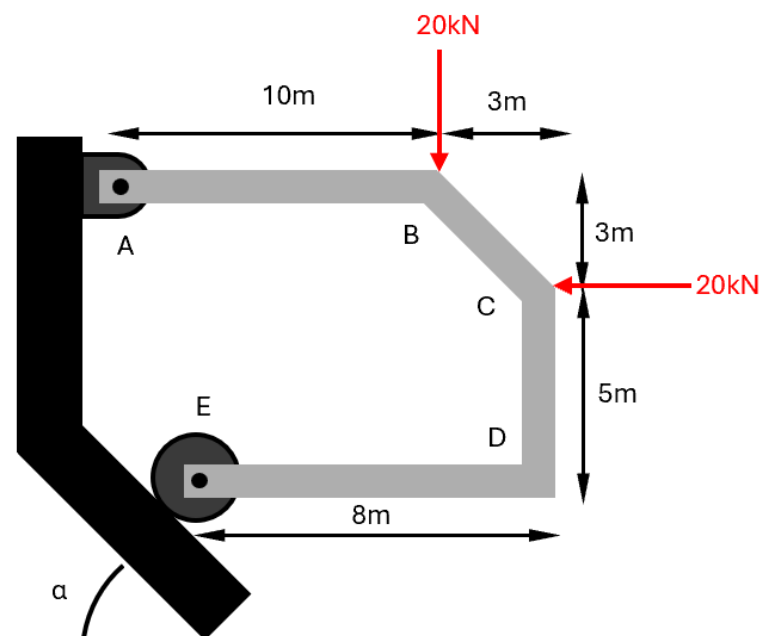
1) Two links AB and DE are connected to a bell crank as shown.



Knowing that the tension in link AB is 720N, determine

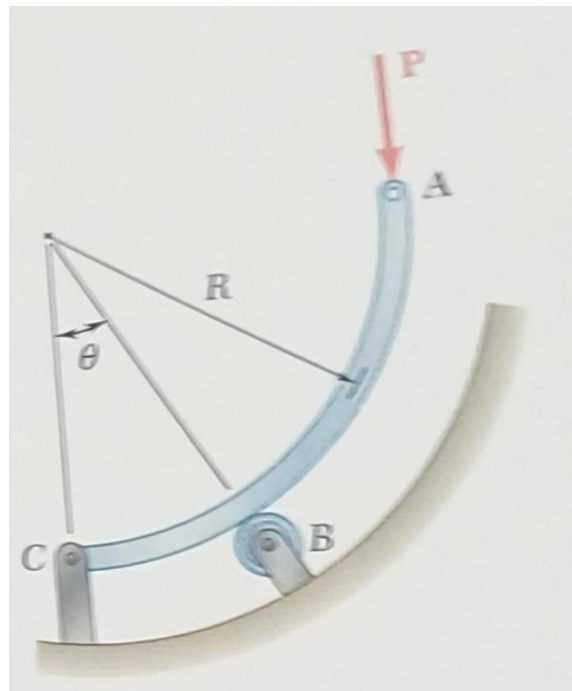
- i) the tension in link DE ii) the reaction at C
- If the maximum allowable value of the reaction at C is 1600N, determine the maximum force that can be safely exerted by AB.

2) For the frame and loading shown, determine the reactions at A and E when



- $\alpha = 30^\circ$
- $\alpha = 45^\circ$

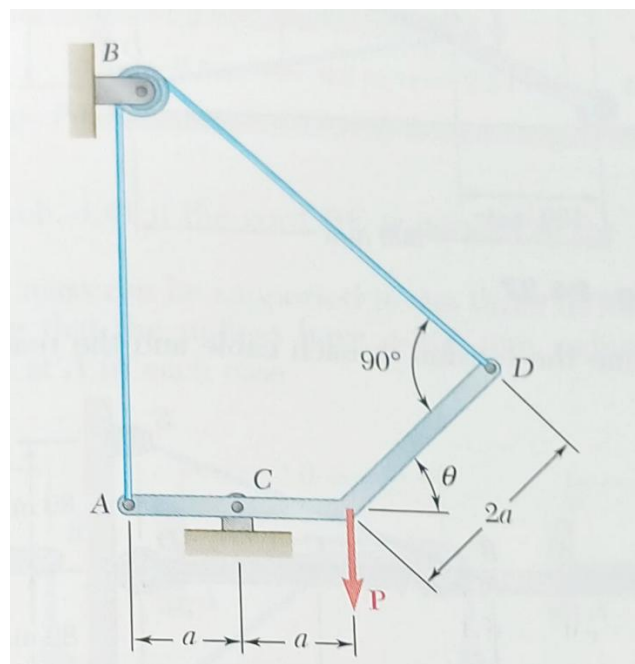
3) ABC is a rod bent into a quarter circle arc of radius R . Assume that A is horizontal to the center of the arc. Find the reactions at B and C given that



a) $\theta = 30^\circ$

b) $\theta = 60^\circ$

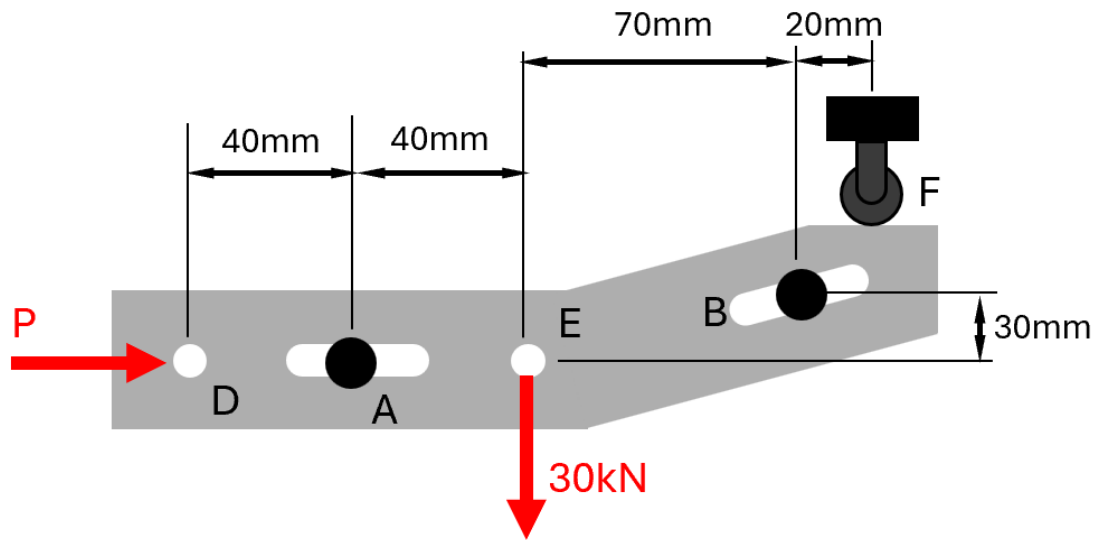
4) Neglecting friction, determine the tension in cable ABD at the reaction at C when



a) $\theta = 60^\circ$

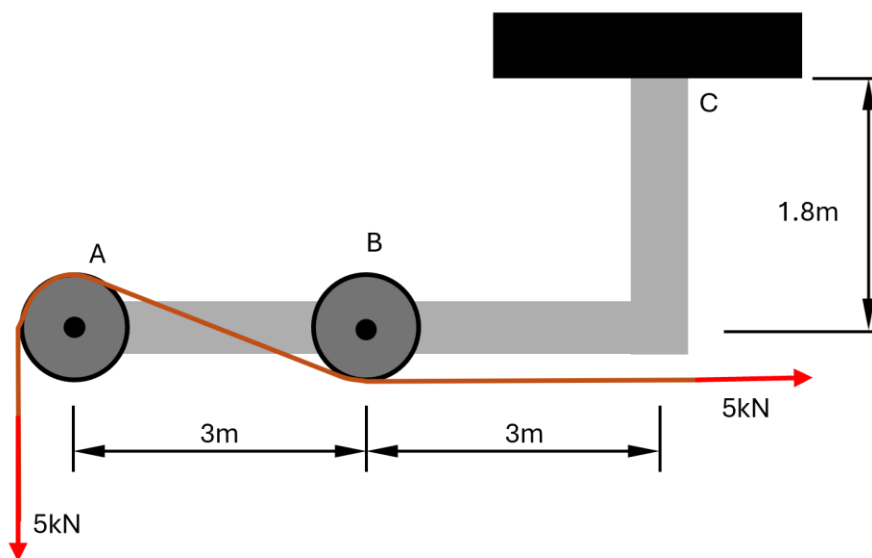
b) $\theta = 45^\circ$

5) Two slots have been cut in plate DEF so that two fixed frictionless pins A and B can be fitted as shown. Given that $P = 15\text{kN}$, determine

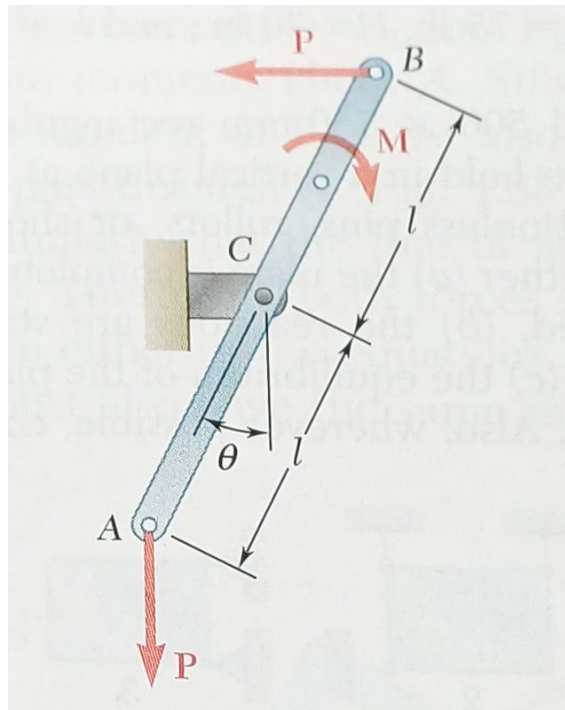


- the force each pin exerts on the plate
- the reaction at F
- the range of values of P if the maximum allowable reaction of F is 20kN .

6) A tension of 5kN is maintained in a cable passing through the support system shown. If the radius of each pulley is 0.1m , determine the reaction at C.



7) Rod AB is acted upon a couple M and two forces of equal magnitude P .

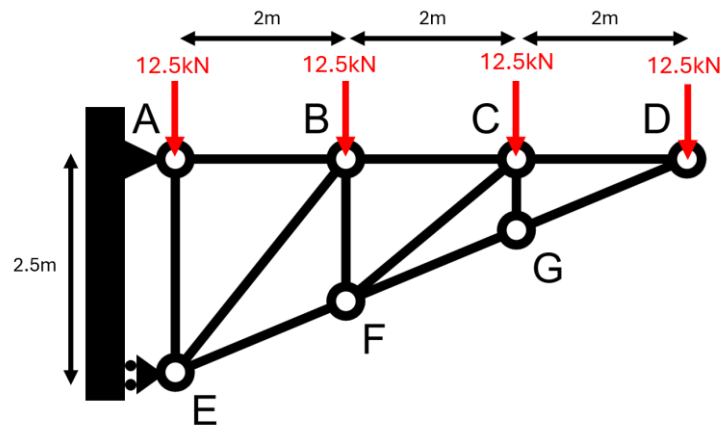


a) Derive an equation in terms of θ , P , M and l that is satisfied when the rod is in equilibrium.

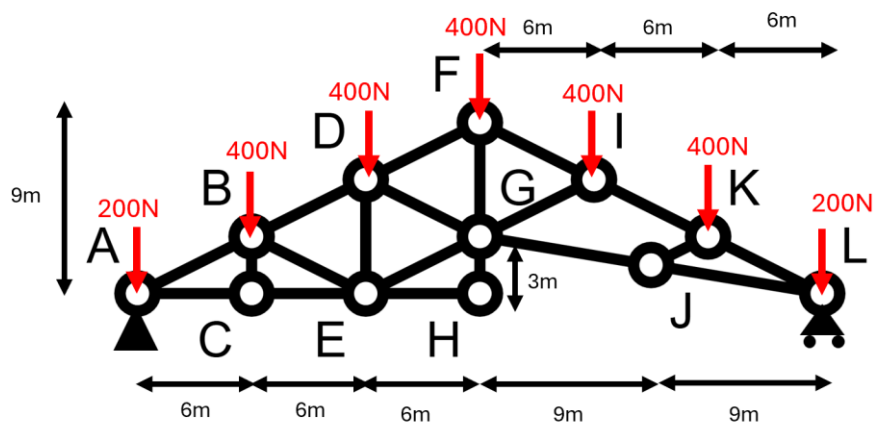
b) Hence, determine θ when $M = 150Nm$, $P = 200N$, $l = 0.6m$

Static Analysis of Trusses

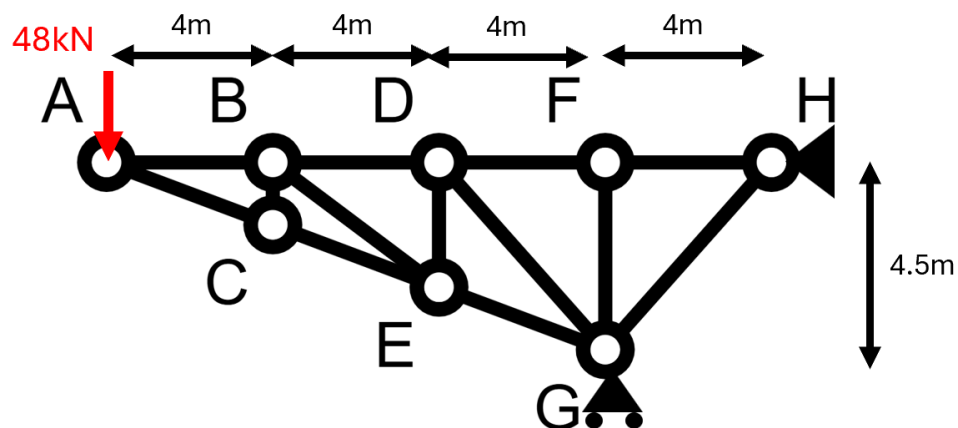
1) Determine the force in each member of the truss as shown.



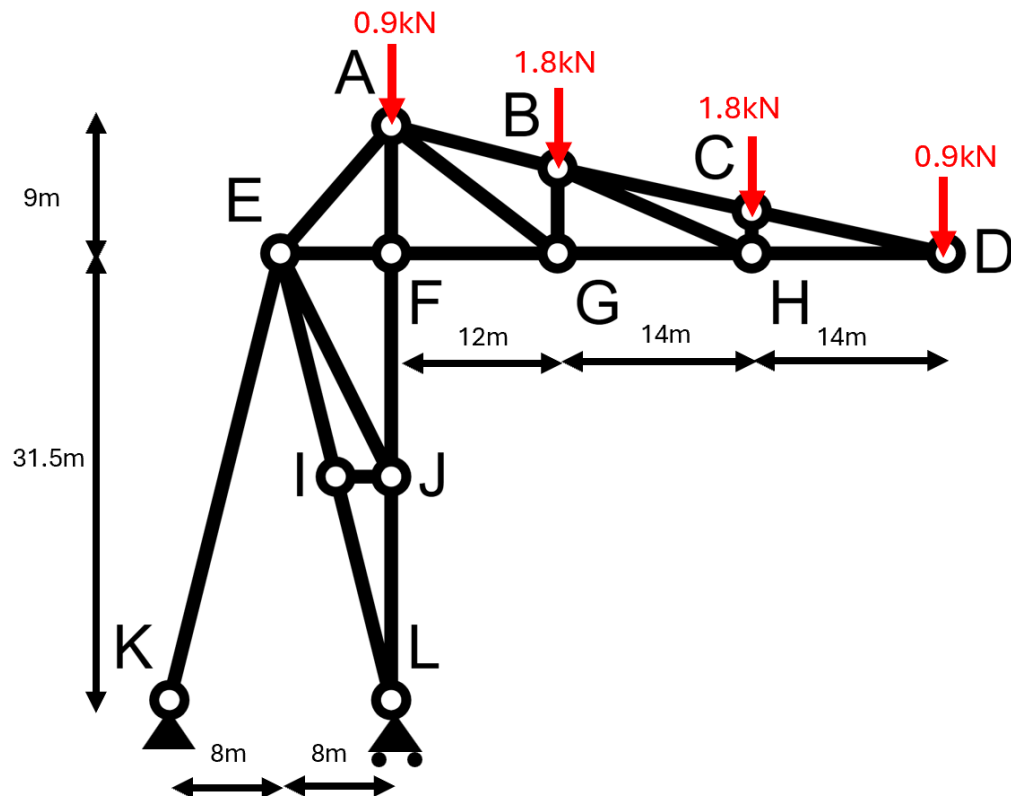
2) For the truss shown, determine the force in FG and in each member on the right of FG.



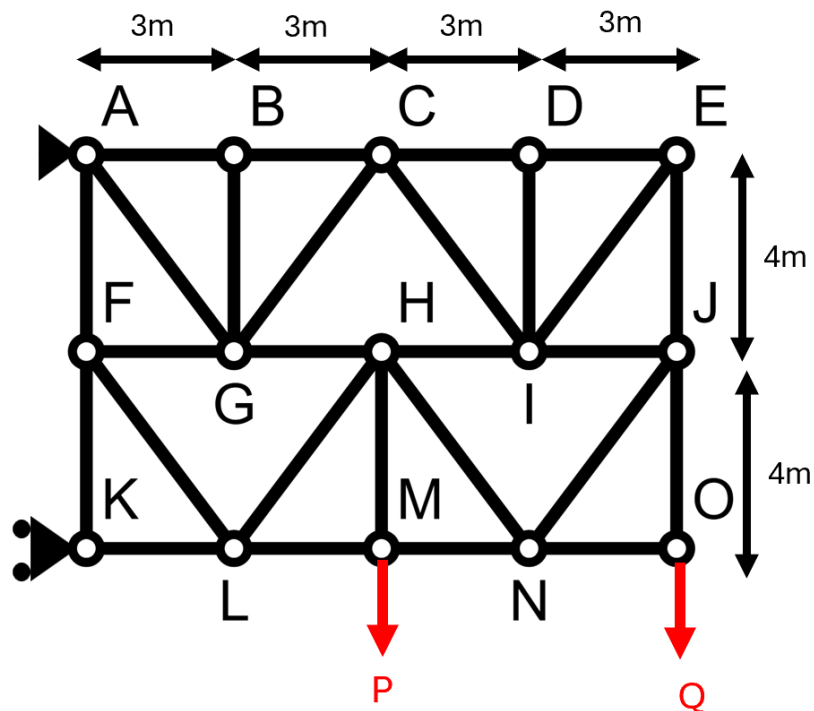
3) Determine the force in each member of the truss shown. Indicate which members are zero force members.



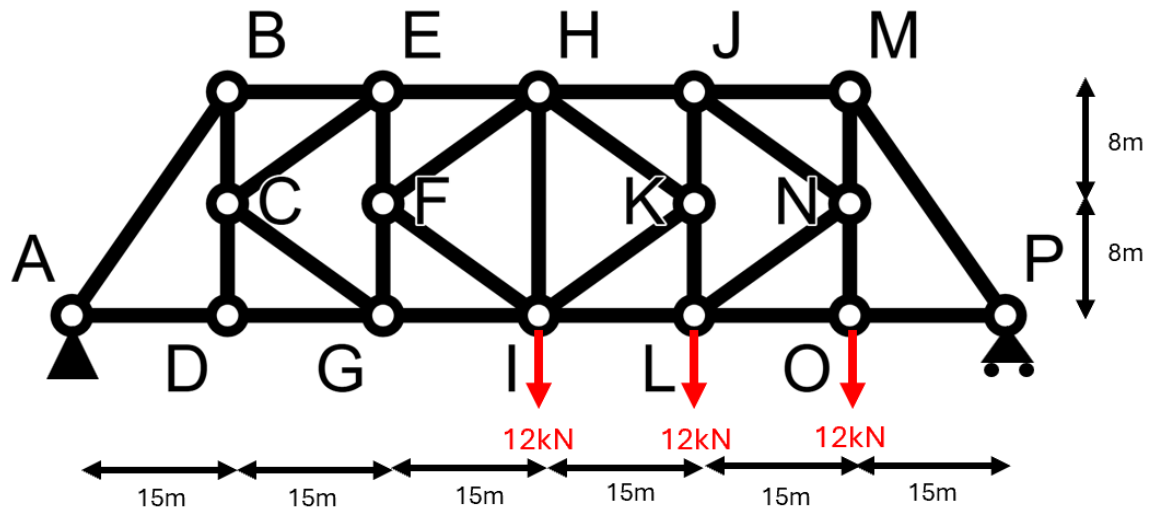
4) Shown is a stadium roof truss, loaded as shown. Determine the forces in members AB, AG, FG, AE, EF and FJ.



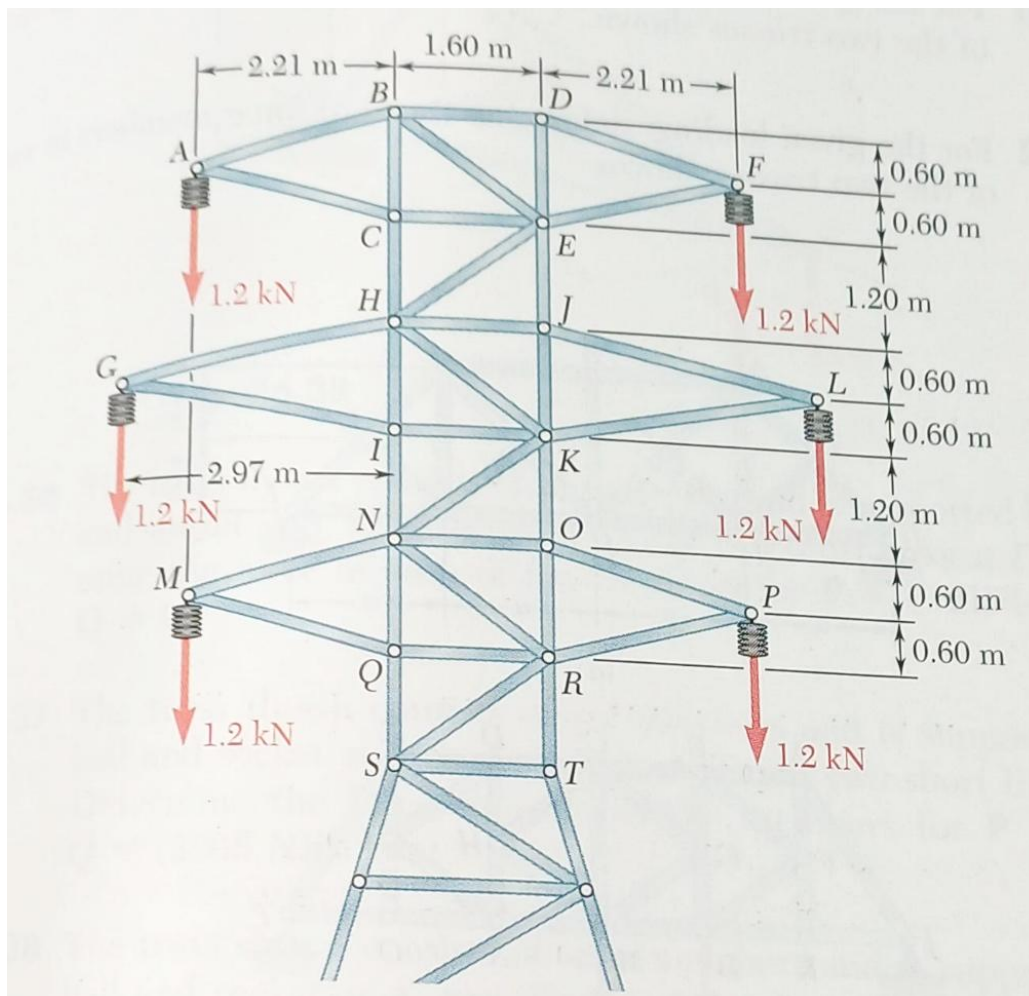
5) For the truss shown, find the forces of members AF and EJ when $P = Q = 1.2kN$.



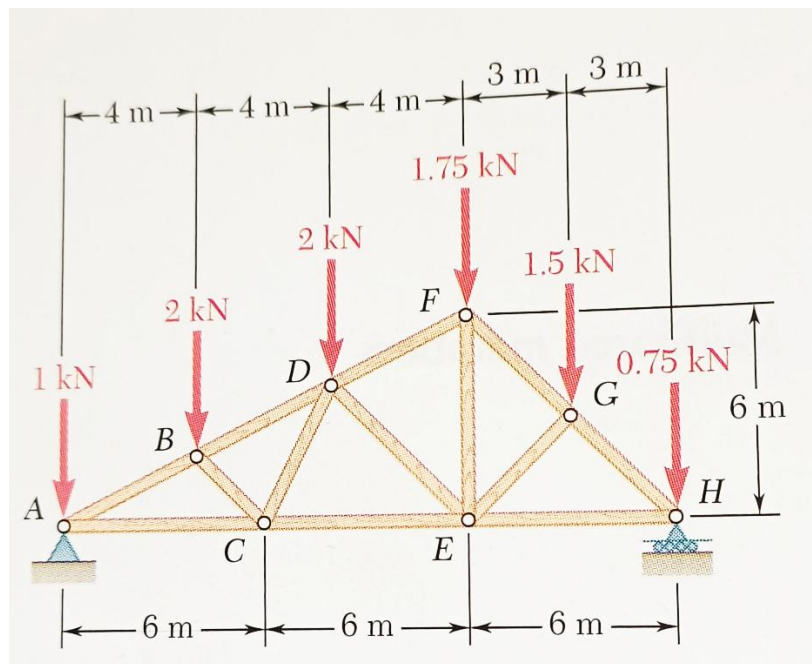
6) For the truss shown, determine the force in members EH, GI, HJ and IL.



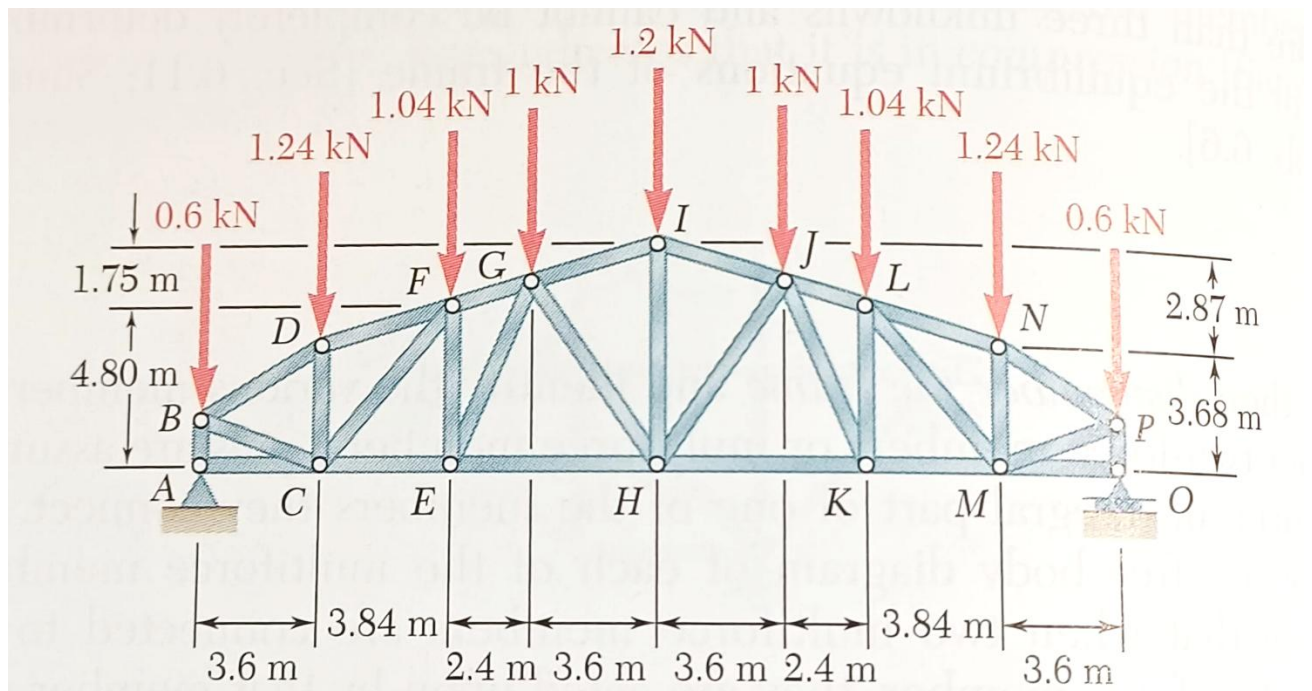
7) The portion of the truss system shown represents the upper part of a power transmission line tower. Determine the force in each of the members above HJ (excluding HJ).



8) Determine the forces in each member of the double pitch roof truss as shown.

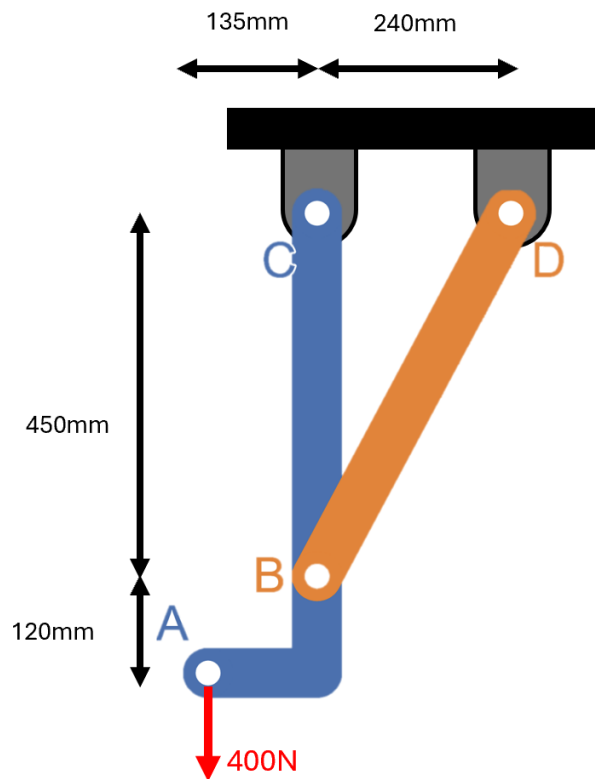


9) Determine the forces in the members FG, EG, EH, KM, LM, and LN of the roof truss shown.

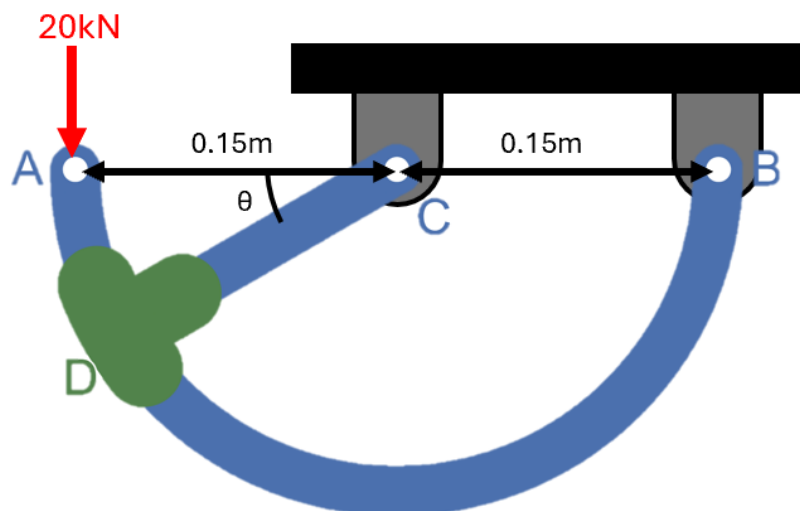


Analysis of Frames and Machines

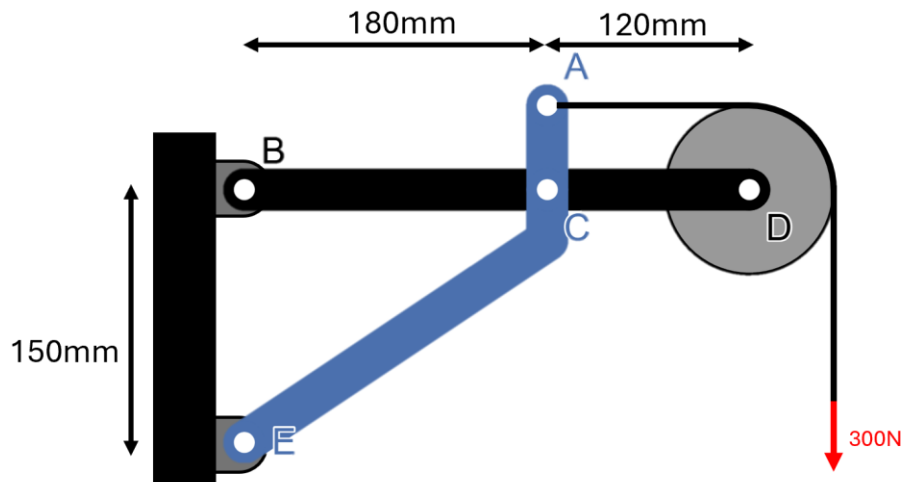
1) Determine the force in member BD and the reaction at C.



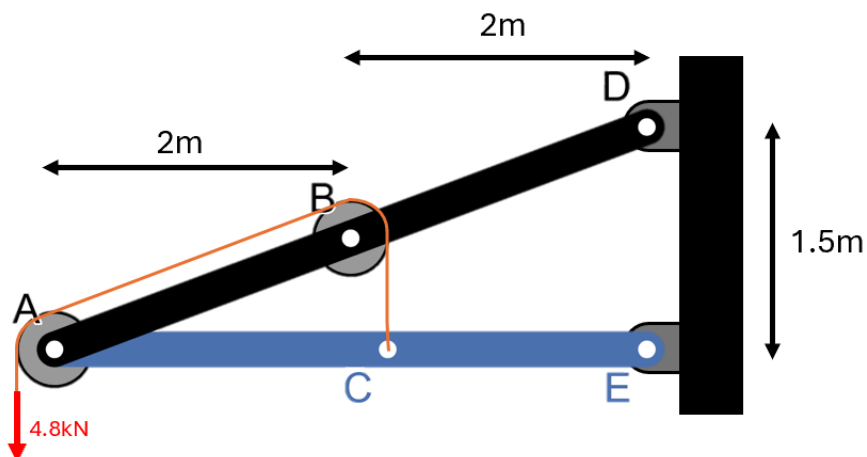
2) Rod CD has a collar at D that can move along rod AB which is an arc of a circle, of center C. If $\theta = 30^\circ$, determine the force in rod CD and the reaction at B.



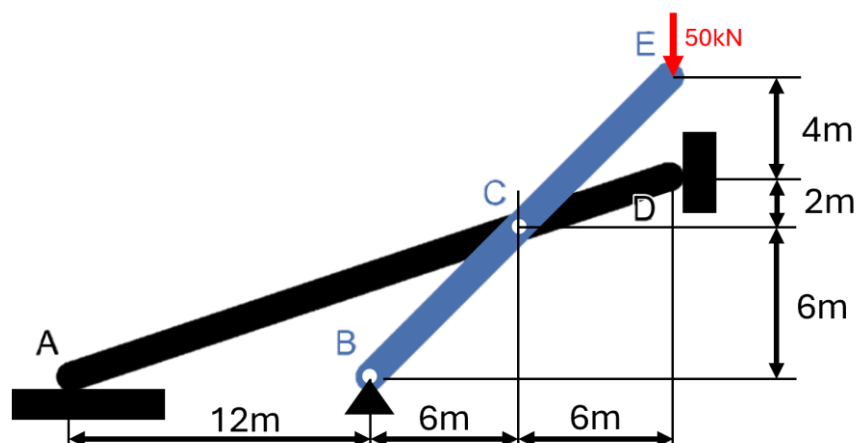
3) Given that the pulley radius is 50mm, determine the components of the reactions at B and E.



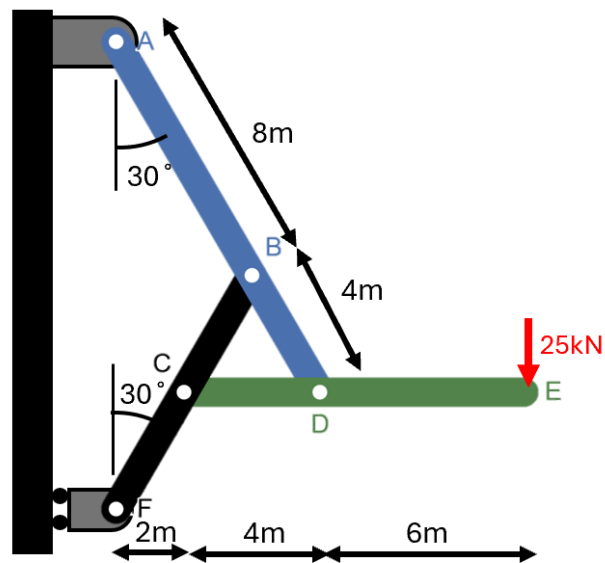
4) Given that the pulley radius is 250mm, determine the components of the reactions at D and E.



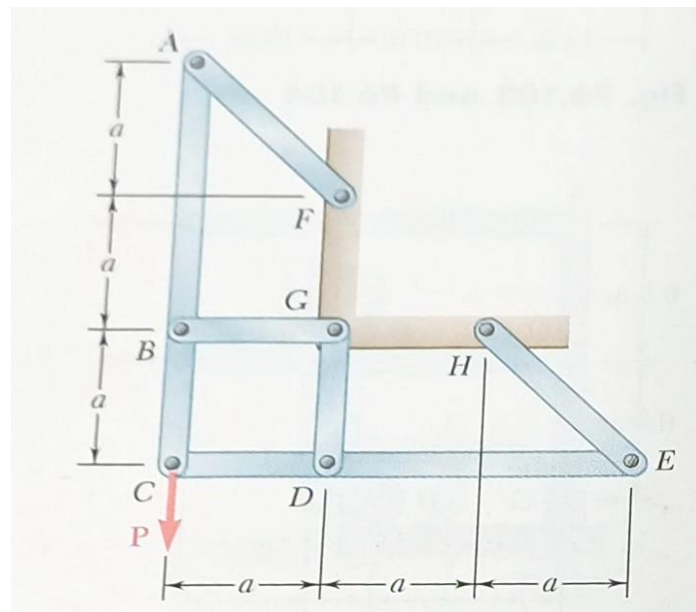
5) Given that the surfaces A and D are frictionless, determine the forces exerted at B and C on member BCE.



6) For the frame and loading shown, determine the forces acting on CDE at C and D.



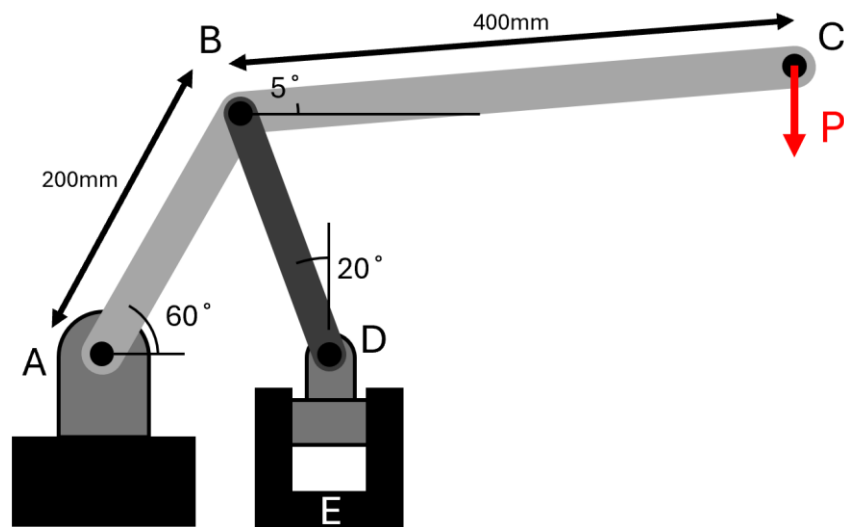
7) Shown is a frame with four links as shown.



a) Determine the force in each **link** of the frame.

b) If force P is replaced by a clockwise couple M_o applied to CDE at D, find the force in each link of the frame.

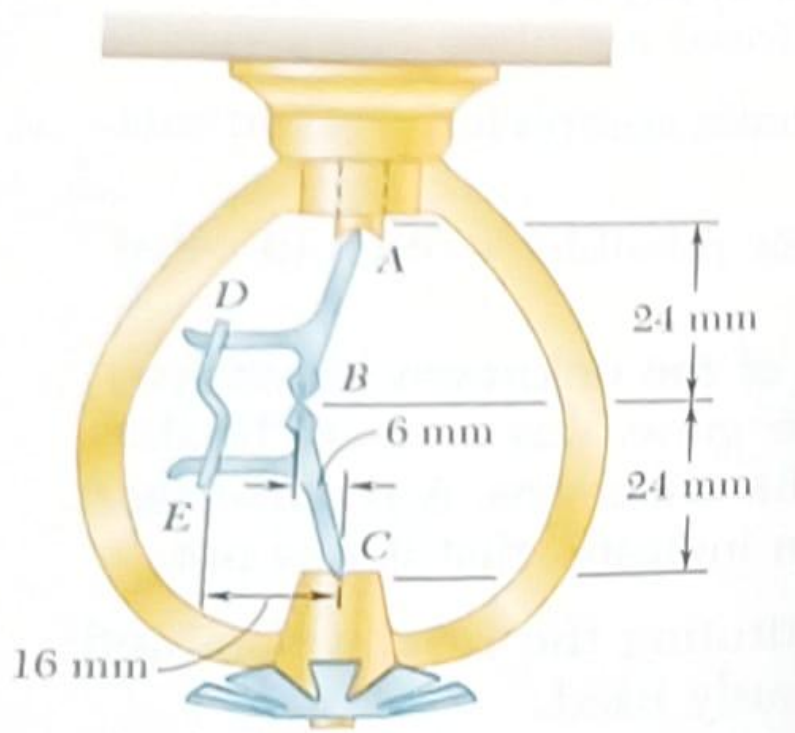
8) The press shown is used to emboss a seal at E.



a) Given that $P = 250\text{N}$, determine the vertical component of the force exerted on the seal, and the reaction at A.

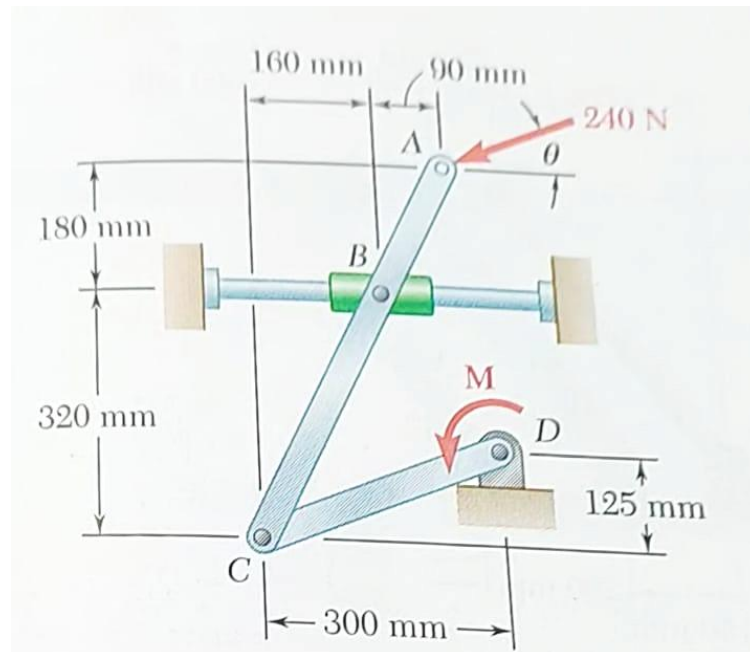
b) Regardless of direction, what is the minimum P needed to have an embossing force of XXXN?

9) A water sprinkler has water pressure exerting 135N downwards on the vertical plug at A. Determine the tension of the fusible link DE and the force exerted on member BCE at B.

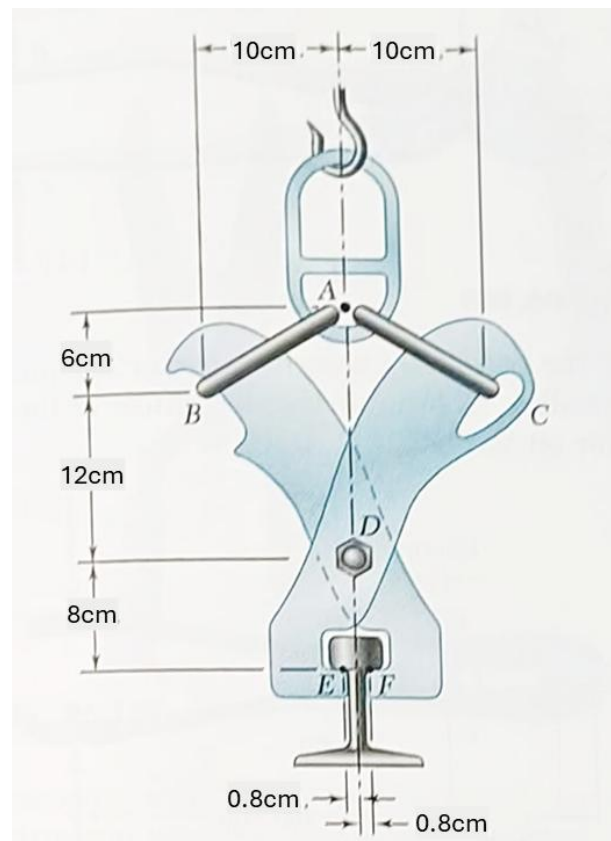


10) Arm ABC is connected by pins to a collar at B and to crank CD at C. Neglecting friction, determine the couple M required to hold the system at equilibrium when

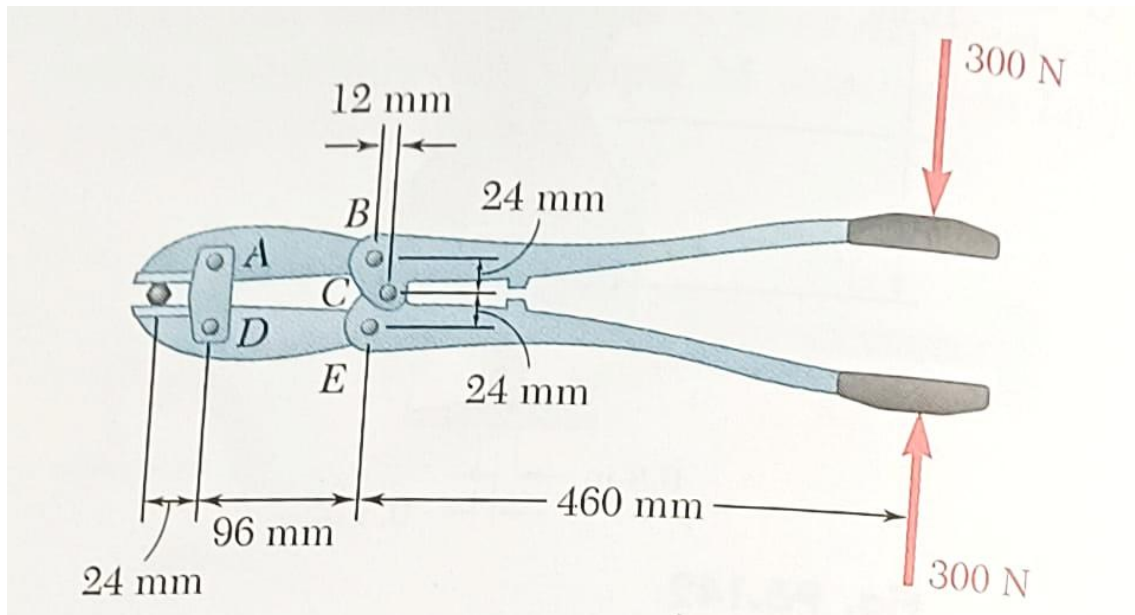
a) $\theta = 0^\circ$ b) $\theta = 90^\circ$ c) $\theta = 30^\circ$



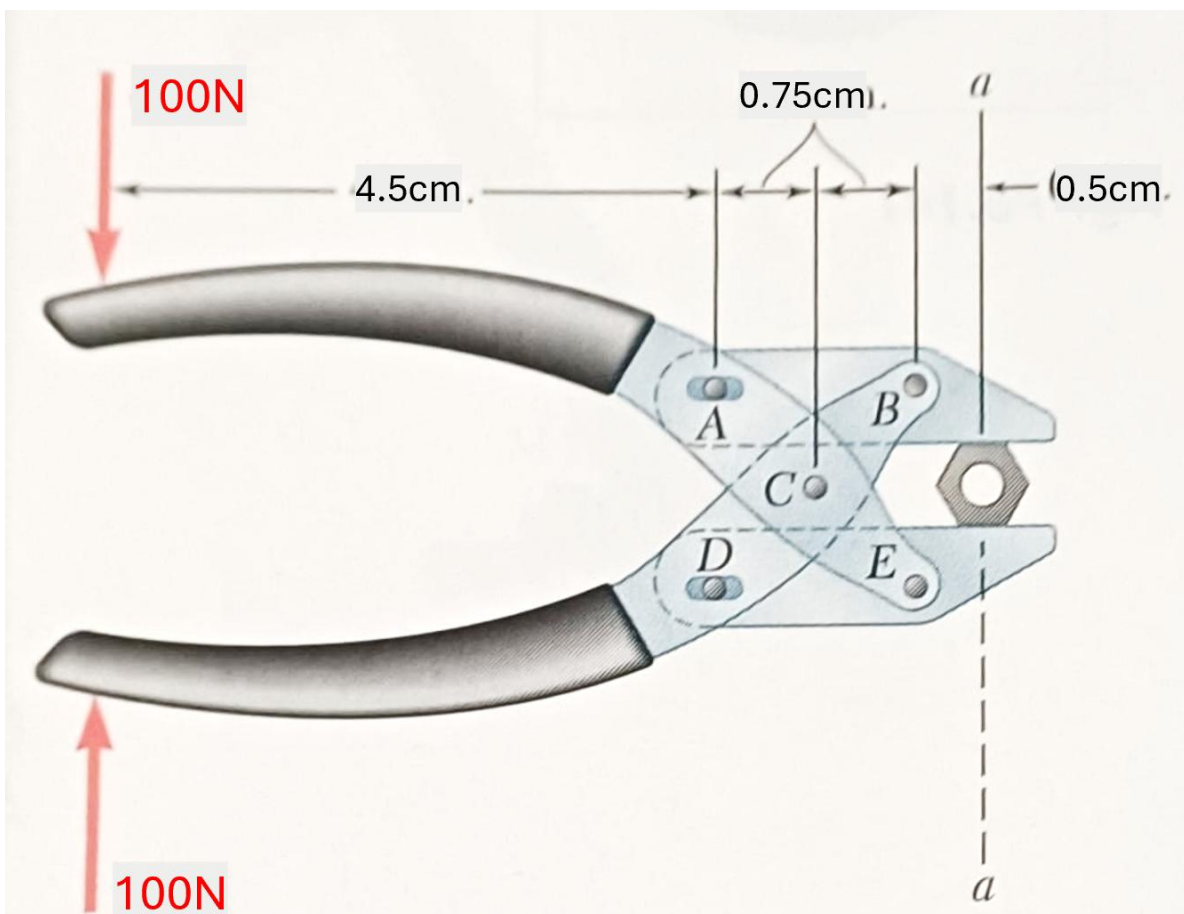
11) A 39m length of railroad rail of weight 44kN/m is lifted by the tongs shown. Determine the forces exerted at D and F on tong BDF. 0.



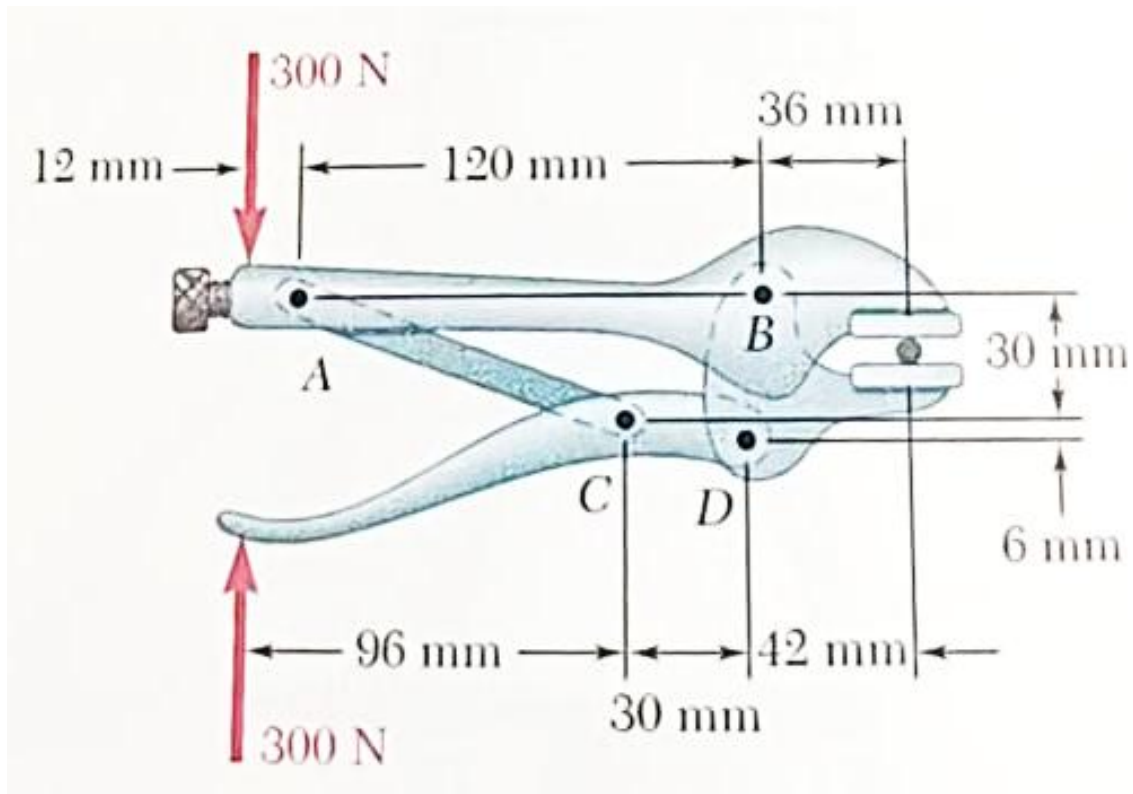
12) Shown is a bolt cutter. Determine the magnitude of the forces on the bolt when two 300N forces are exerted on the handles.



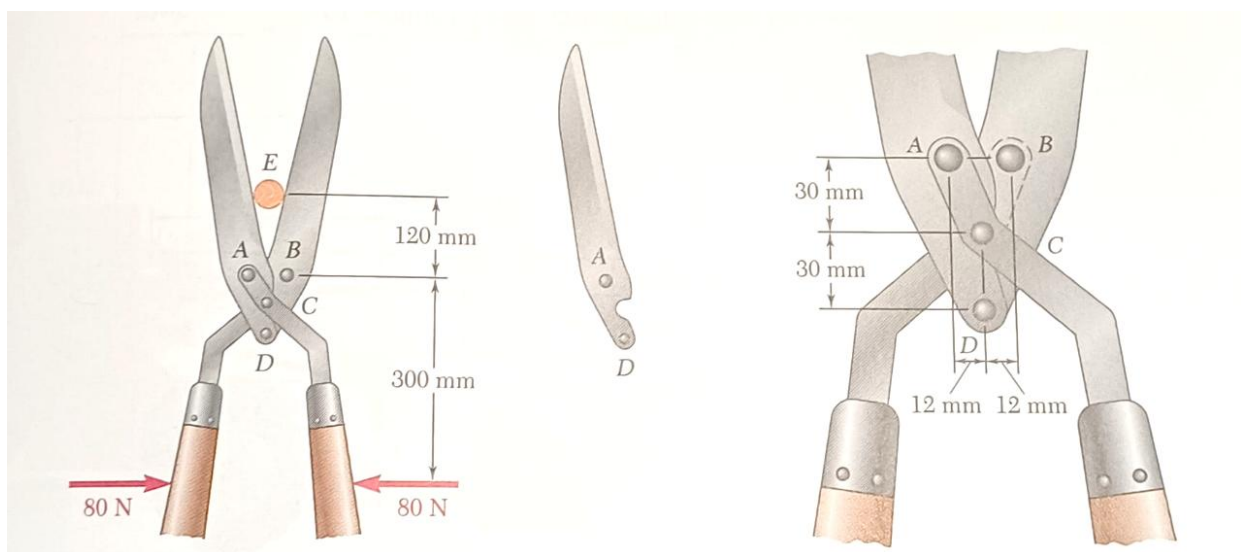
13) Determine the magnitude of the gripping forces exerted along line aa on the nut when two 100N forces are applied to the handles as shown. Assume pins A and D are frictionless pins.



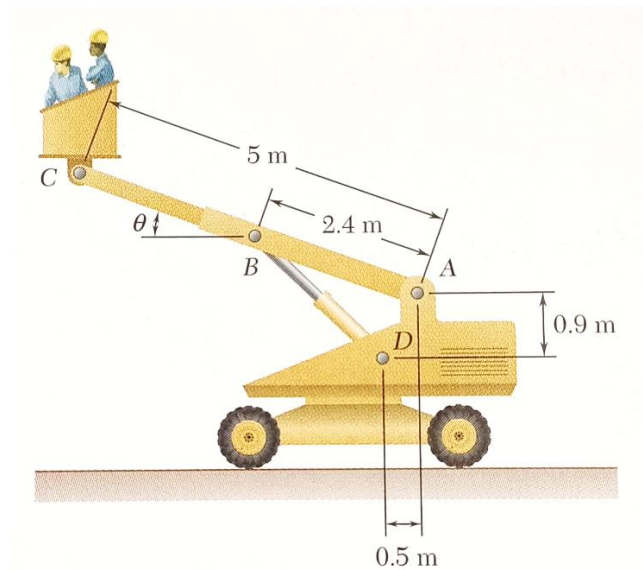
14) Determine the magnitude of the gripping forces produced when two 300N forces are applied as shown.



15) Shown is a pair of garden shears, with two blades and two handles. The two handles are connected by pin C and the two blades are connected by pin D. The left blade and the right handle are connected by pin A; the right blade and the left handle are connected by pin B. Determine the magnitude of the forces exerted on the small branch at E when two 80N forces are applied to the handles.



16) Shown is a telescoping arm ABC supporting construction workers on a platform, of combined mass 200kg.



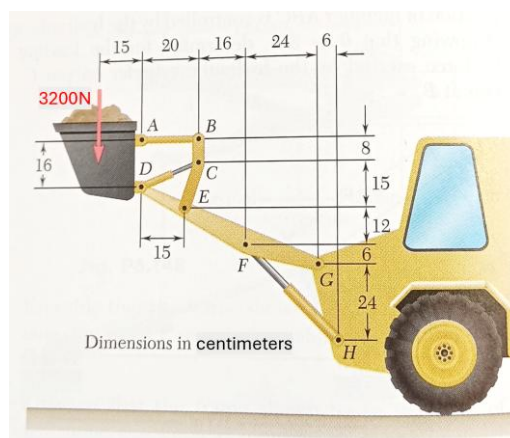
a) When lifted, $\theta = 20^\circ$. Determine

- the force exerted at B by the single hydraulic cylinder BD
- the force exerted on the supporting carriage at A.

b) When the arm is lowered, $\theta = -20^\circ$. Determine

- the force exerted at B by the single hydraulic cylinder BD
- the force exerted on the supporting carriage at A.

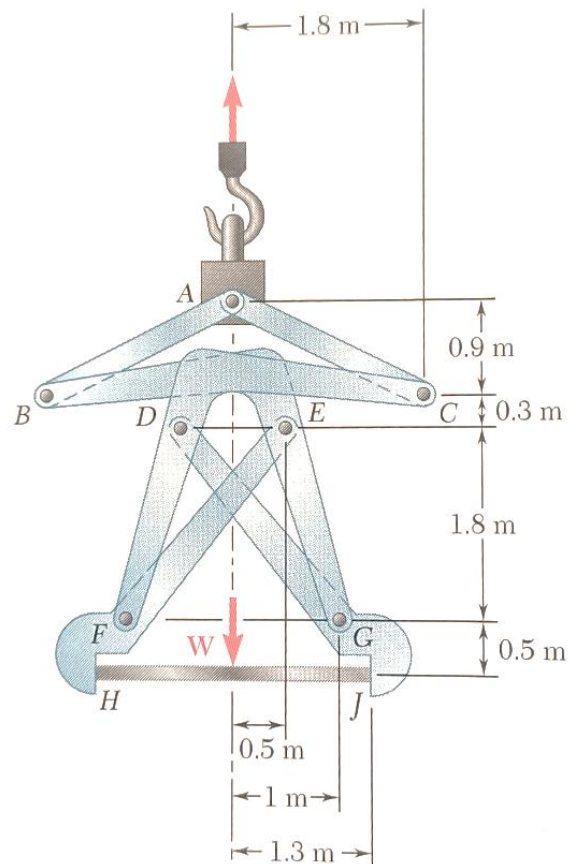
17) The bucket of a front-end loader shown carries a 3200N load. The motion of the bucket is controlled by two identical mechanisms, one of which is shown. Knowing the mechanism shown only supports one half of the load, determine the force exerted by



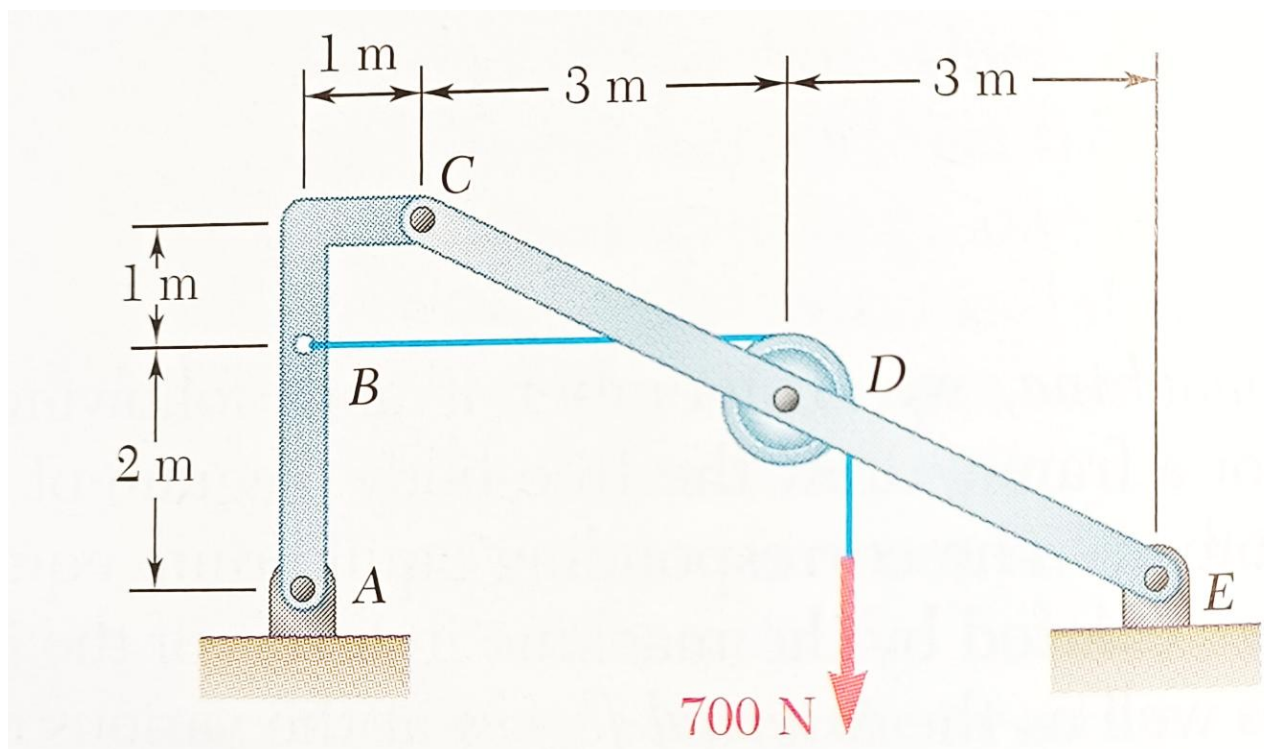
a) cylinder CD

b) cylinder FH

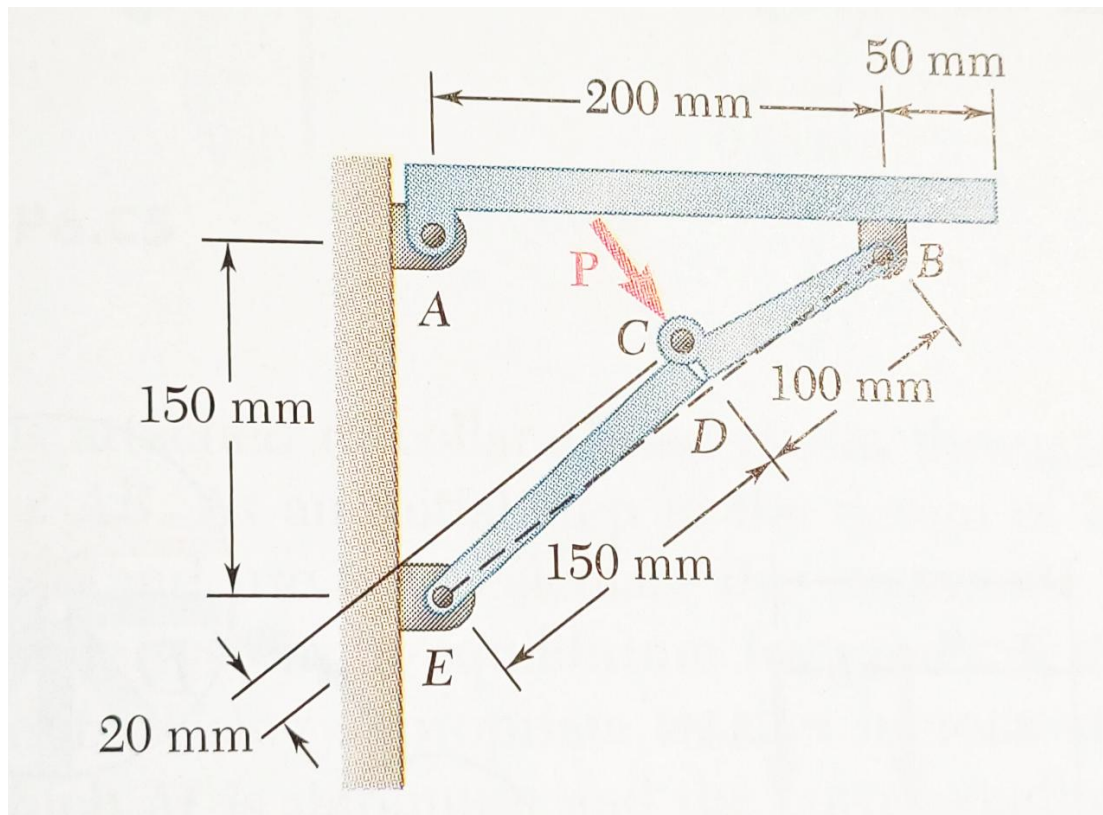
18) The tongs shown are used to grab and lift a 7500kg steel slab, HJ. Assuming slipping does not occur, determine all forces acting on member EFH.



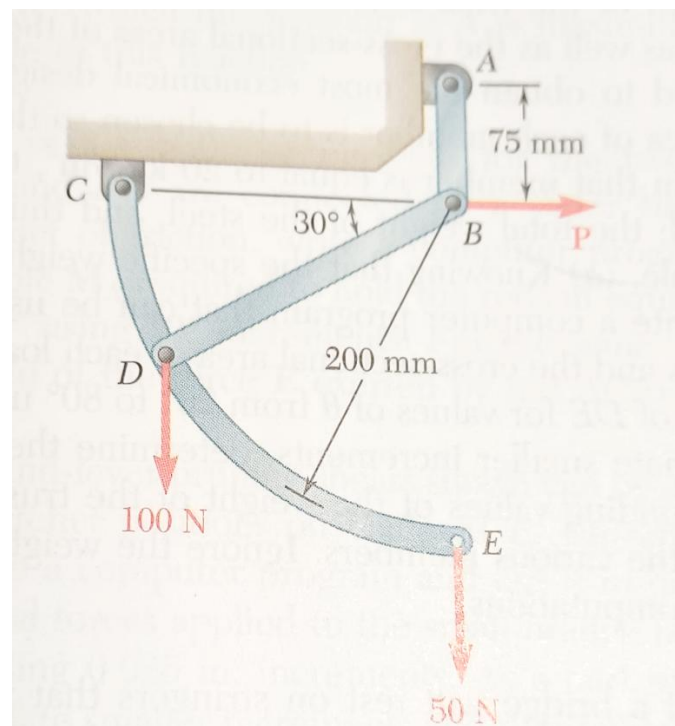
19) Find the reactions A and E of the frame, given that the pulley has diameter 0.5m.



20) A 20 kg shelf is held horizontally by a self-locking brace that consists of two parts EDC and CDB hinged at C. Determine the force P needed to release the brace. Assume the mass of the shelf acts at its centroid.



21) Shown is a frame where CDE is a circular arc. Determine
a) force P b) corresponding force at BD c) reaction at C



Answers

Vector Forces & Moments

1) $T_{CA} = 5492.5N, T_{CB} = 509.43N$

2a) $P = 601.7N, \alpha = 46.83^\circ$ 2b) $P = 630.22N, \beta = 22.5^\circ$

3a) $M_C = 33.425Nm$ (acw), $F_C = 160N, 60^\circ$ 3b) $B = -20N, D = 177.6N, 63.23^\circ$

4) $x = 50mm, -900N$

5) $\alpha = 30^\circ, P = 300N$

6a) $54.92mm$ below F 6b) $312.9mm$ rightward of H

7a) $M_D = \frac{2bFx}{\sqrt{a^4+4b^2x^2}}$ 7b) $\frac{4}{\sqrt{17}}F$

Static Rigid Bodies

1ai) $600N$ 1aii) $1252.8N, 69.82^\circ$ b) $919.54N$

2a) $R_E = 45.36kN, R_A = 19.47kN, -97.91^\circ$ 2b) $R_E = 36.77kN, R_A = 8.485kN, -135^\circ$

3a) $R_B = 2P, C_x = P, C_y = P(1 - \sqrt{3})$ 3b) $R_B = \frac{2P}{\sqrt{3}}, C_x = P, C_y = P(1 - \frac{\sqrt{3}}{3})$ (might be wrong?)

4a) $T = \frac{2}{3}P, C_x = \frac{\sqrt{3}}{3}P, C_y = 0$ 4b) $T = \frac{2}{2+\sqrt{2}}P, C_x = \frac{\sqrt{2}}{\sqrt{2}+2}P, C_y = 0$ (might be wrong?)

5a) $R_B = 38.08kN, R_A = 18.84kN$ 5b) $R_F = 23.84kN$ 5c) $P \in [0, 14.74kN]$

6) $C_x = 5kN, C_y = 5kN, M_C = 39.5kNm$

7a) $Pl(\sin(\theta) + \cos(\theta)) = M$ 7b) 17.11°

Static Analysis of Trusses (+ve means tension, -ve is compression)

1)

Member	AB	BC	CD	AE	BF	CG
F/kN	60	45	30	37.5	6.25	0
Member	BE	FC	EF	FG	GD	
F/kN	-24.01	-19.53	-48.75	-32.5	3.253	

2)

Member	FG	FI	IK	KL
F/kN	-1.867	1.64	1.193	1.193
Member	GI	GJ	JK	JL
F/kN	-0.4472	1.108	0	1.108

3) $F_{BC} = F_{BE} = F_{ED} = F_{DG} = F_{FG} = 0kN$ (zero force members)

$$F_{AB} = F_{BD} = F_{DF} = F_{FH} = 128kN$$

$$F_{AC} = F_{CE} = F_{EG} = -136.7kN$$

$$F_{HG} = 194.6kN$$

4)

Member	AB	AG	FG
F/kN	8.2	4.5	-11.6
Member	AE	EF	FJ
F/kN	17.46	-11.6	-6.147

5) $F_{AF} = 1.5kN, F_{EJ} = 0.9kN$

6) $F_{EH} = -22.5kN, F_{GI} = 22.5kN, F_{HJ} = -33.75kN, F_{JL} = 33.75kN$

7)

Member	AB	BD	DF	BC	BE	DE
F/kN	2.29	2.21	2.29	-0.6	0	-0.6
Member	AC	CE	EF	CH	EH	EJ
F/kN	-2.29	-2.21	-2.29	-1.2	0	-1.2

8)

Member	AB	BC	AC	BD	CD	CE	DE
F/kN	-7.826	1.886	7	-9.317	-2.386	8.774	1.508
Member	DF	EF	FG	EG	GH	EH	
F/kN	-11.7	13.95	-14.8	9.497	-5.303	-3.75	

9)

Member	EH	FG	EG
F/kN	5.081	-5.231	-0.1476
Member	LN	LM	KM
F/kN	3.954	-1.963	5.022

Analysis of Frames and Machines

1) $F_{BD} = 255N(C), C_x = 120N, C_y = 625N$

2)

3)

4)

5)

6)

7)

8)

9)

10)

11)

12)

13)

14)

15)

16)

17)

18)

19)

20)

21)